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The Effects of Topology, Gender, and Task Type on Web Browsing Performance in Senior Citizens

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Abstract

Searching for information on websites that have complicated layouts is difficult and may result in errors and frustration. There is a particular need to develop and test web designs for senior citizens, who may have operational difficulties that affect such searches. Objectives of the present study were to 1. identify correlations among the effects of topology, gender, and task type on web browsing tasks in senior citizens, and 2. design and compare different website design examples for senior citizens. The current study involved 51 senior citizens (26 females, 25 males) between the ages of 61 and 71 who were recruited for hypertext browsing tests. A 2 × 2 × 3 mixed factorial design was used; between-subjects factors were topological structure (2 levels: linear and mixed) and gender (male and female), and the within-subjects factor was task type (3 levels: easy, moderate, and difficult). The results yielded two two-way interactive effects. First, the interaction between topological structure and task type was significant. A mixed structure led to superior performance compared with a linear structure under the moderate and difficult task conditions, but no difference was observed in the easy task. Second, the interaction between gender and task type was also significant. As with the other two-way interaction, no difference was observed under the easy task condition, but men took less time than women to conduct the search under the moderate and difficult task conditions. This study extends the findings of previous research by demonstrating interactive effects involving gender, topological structure, and task type on hypertext browsing performance in senior citizens. Results of the study may be of value for web page design, and they suggest that variables such as gender, topological structure, and task type should be considered when targeting older populations.

Keywords: Website Design, Senior Citizens, Task Performance

1. Introduction

According to a report by the Taiwanese Government, in September 2016, 14.6% of Taiwan's citizens would reach the age of 65 years, with an 'aged society' predicted to be in place before 2018 (DGBAS, 2016). As argued in a previous study, the design of appropriate interfaces may encourage senior citizens to use digital technologies (Sayago & Blat, 2010); however, operational difficulties resulting from the physical and psychological effects of aging and their impact on senior citizens while browsing the web for information have yet to be investigated (Xie, 2003). One major limiting factor in the navigation of hyperlinks is that users may become disoriented

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as a result of complex links (Lin, 2004). Improved interface designs for senior citizens are in demand (Birkeland, 2009; Melenhorst et al., 2006). Previous studies have addressed issues such as topological structure, task characteristics, or individual differences (McDaniel et al., 2000, Chang et al., 2011) identifying important factors for improving the efficiency of web browsing performance. However, little research has been done to identify the issues specific to senior citizens as they conduct web browsing. In the present study, we recruited senior citizens to participate in hypertext browsing tests related to the website of a company selling electric wheelchairs to senior citizens, and examined various factors, such as topology, gender, and task type, and their correlation with web browsing performance.

There are three general topological categories of hyperlink structural design: linear, hierarchical, and network structures (Lin, 2004). A linear topology is characterized by linear links, resulting in lower demands on working memory, but it uses a rigid structure in which tedious backward and forward moves may consume more time relative to other designs, particularly when the search task is complex. A hierarchical topology relies on parent-child links, with all links layered in a hierarchical structure; this may reduce unrelated link moves, but users may become lost within the hierarchical structure when the task requires them to switch between layers. A network structure visually represents all nodes, icons and critical links within the same display, but may result in users becoming distracted by the complex visual information and ultimately spending more time searching through the target web page.

In a previous study examining performance, hierarchical and network topologies were not significantly different, suggesting that a mixed topology may integrate the merits of these two topologies (Chang & Chien, 2015). It is important to note that there is no existing standard structure for mixed designs; repeated testing and modification of the interface are necessary during development. Additionally, the comparison between linear and mixed topologies remains to be tested.

Lin's (2004) study examining the topology of website design reported that senior citizens performed better with a linear topological interface design than with hierarchal and network designs. Others have suggested that a consideration of the interaction between task complexity and cognitive factors is critical in assessing wayfinding task performance (Vidal & Berthoz, 2005). A multi-link hyperlink task, which requires cross-referencing information from more than one web page, is similar to the representation of a network topology, but hypothetically involves less search time and a smaller number of pages that need to be opened. In contrast, a single-node task is similar in structure to a linear topology; thus, a lower memory load may be imposed by a single-node task, potentially resulting in greater efficiency. Previous research also indicates differences in performance between men and women on wayfinding tasks (Coluccia & Losue, 2004; Lawton & Morrin, 1999), with one study reporting a significant interaction between task type of complexity and gender with respect to web browsing performance (Chang & Chien, 2015). The topological structure of the website, gender differences, and task complexity affect website task performance, but have received little attention in research on web browsing by senior citizens. The objectives of the present study were to (1) identify relationships of topology, gender, and task type with web browsing in senior citizens, and (2) design and compare different website design examples for senior citizens.

In the present study, we hypothesized the following patterns of topological effects among older users:

H1:Topological structure significantly affects task performance by elders during web browsing.

H2:Gender, task type, and topological structure have interactive effects on the performance of senior citizens during web browsing.

2. Materials and design

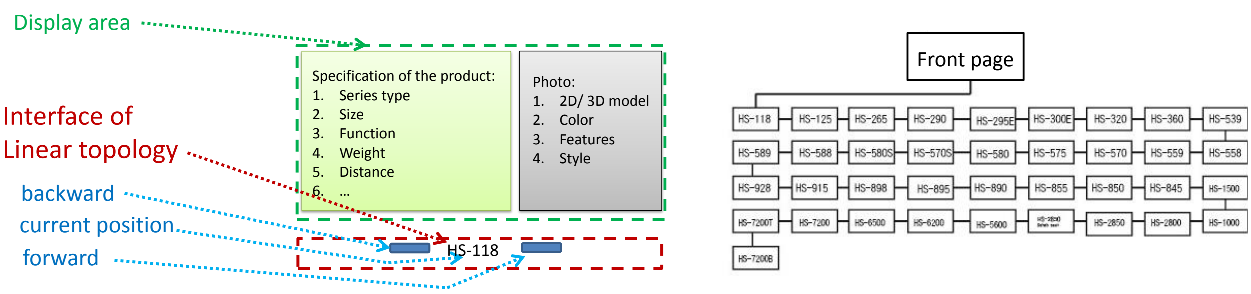
2.1 Participants

Fifty-one participants (25 females, 26 males) aged between 61 and 71 years with a mean age of 64.5 (SD = 3.03) years, all of whom passed a standard visual test of web page browsing proficiency, were invited to participate in hypertext browsing tests. All of the participants were retired people who were recruited from the Activity Centre of the Elderly, North Branch of Taoyuan County, Taiwan, and who had daily experience with internet browsing on a computer. Participants were randomly assigned to one of the topological structure conditions. All participants were paid US $10 per hour for their participation.

2.2 Topological structure

2.2.1 Linear topology

Figure 1 shows the layout design (Fig.1(a)) and link structure (Fig.1(b)) of the linear topological web page design. There were two main issues in the layout design: display area and interface. The display area includes specifications and photos of the product displayed. The interface includes the operational links for forward, current, and backward functions (Fig.1(a)). The linear topological web page design featured each node in a fixed linear sequence. Each individual web page could be accessed by clicking on the forward or back button at the bottom of each page, which caused the next or previous page to be displayed. The linear links were based on one parent node for every two child nodes, resulting in 38 nodes with 37 links.

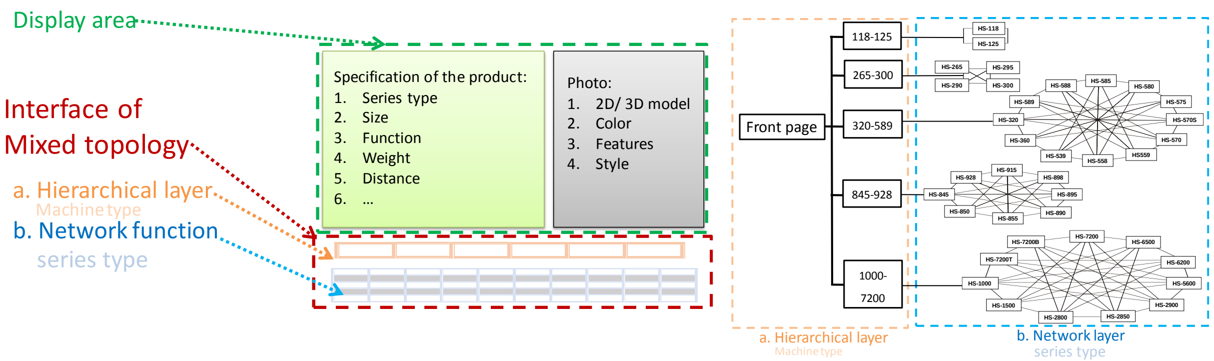


(a) Layout design (b) Link structure

Figure 1. Linear topological web page design

2.2.2 Mixed topology

Figure 2 shows the layout design (Fig.2(a)) and link structure (Fig.2(b)) of the mixed topological web page design. Similar to the linear topology, the specification and photos of the product are displayed in the display area. Unlike the linear topology, the interface of the mixed topological design consisted of two types of linking system, hierarchical and networked layers; the latter allowed each page to be accessed by clicking on its name in a table in the text. In the hierarchical layer, five main product categories of electric vehicle, categorized by size and labeled with series numbers, triggered a corresponding network layer (Fig.2(a)). In the network layer, all function nodes were connected, with cross-referenced links to connect every web page with every other page.



(a) Layout design (b) Link structure

Figure 2. Mixed topological web page design

2.3 Task type

Two tasks for each task type (single-layer, two-layer, and three-layer types) were developed, resulting in six tasks in all (Table 1). The specific task types were as follows:

(1) Easy: the task was to navigate through a single linear structure involving five linking nodes to find a specific page that provided the answers to given questions. In Task No. 1, as an example (Table 1), the subject was required to identify the safe weight of the series type HS-265. From the required linking node to the destination, five link nodes were needed to reach the target answer based on the linear topological structure.

(2) Moderate: the task was to navigate through two targeted missions involving 25 linking nodes to find two specific pages that provided answers to given questions. In Task No. 3, as an example (Table 1), the subject was required to identify the color of the series type HS-125 and HS-570S. HS-125 is in the Micro category, whereas HS-570S is in the Medium category. Thus, two categories of product layer were involved, and 25 linking nodes were needed to reach the destination, based on the linear topological structure.

(3) Difficult: the task was to navigate through three targeted missions involving 45 linking nodes to find three specific pages that provided answers to given questions. In Task No. 5, as an example (Table 1), the subject was required to identify the

numbers of the car wheel for the HS-300, HS-588, and HS-890. HS-300 was in the Mini category, HS-588 in Medium, and HS-928 in Max. Thus, three categories of product layer were involved, and 45 linking nodes were needed to reach the destination, based on the linear topological structure.

Table 1. Types and levels of task design

No.	level	Task	Number of nodes
Task 1	easy	Single mission	5
Task 2	easy		
Task 3	moderate	Double mission	25
Task 4	moderate		
Task 5	difficult	Treble mission	45
Task 6	difficult		

2.4 Experimental design

A mixed-factor 2 × 2 × 3 MANOVA experimental design was employed in this study. The independent variables were gender, hyperlink topology (linear and mixed), and task type (easy, moderate, and difficult). In the present study, the definition of web browsing performance was the mean time taken to locate specific nodes. A within-subjects design was used for task type, and a between-subjects design was adopted for gender and topology. The experiment consisted of a total of six tasks, with two tasks for each of the three types. To avoid a learning effect and the use of different starting nodes for each task, the tasks were arranged using a Latin square, and each task started with the first page of the website.

There are limitations in generalizability due to this experimental design. Other dependent variables related to web browsing performance, wrong turns, and correct rates were confounded by task complexity and topology effects (Coluccia & Louse, 2004). First, it was observed during the experiment that backtracking frequency and search time varied by task complexity; we did not analyze backtracking frequency. Second, unlike the linear topological structure, in a mixed topology condition with neither a correct route nor direction to be followed, the statistics for wrong turns were confounded; we did not count the correct rates.

2.5 Procedure

All documents were run on Microsoft’s Internet Explorer 9.0 browser and were displayed on a 15.6-in LCD color monitor. The screen size of the projection area was about 360 × 210 cm. Participants were asked to sit in front of the display at a distance of 50–65 cm (Fig. 3). The resolution was 1024 × 768 pixels, and the frame rate was 85 Hz. Participants’ activity was monitored throughout the experiment.

The participants were instructed to learn to interact with the user interface by performing practice tasks. No time constraints were imposed during the practice session. The session ended when the participants were confident about their basic understanding of the interface hierarchy and the relationships among the various nodes.

After the participants had familiarized themselves with the web design, the experiment began. Each task was explained on a task card (Fig. 4), and participants followed the requirements specified to complete the navigational tasks. Participants were required to complete all of the interaction tasks. The full experiment lasted approximately 1–1.5 h for each participant.

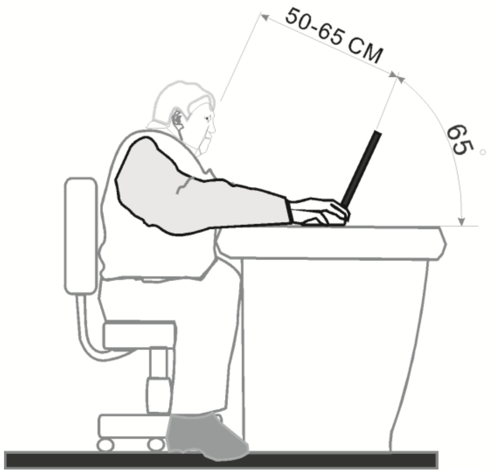


Figure 3. The situated participant and display

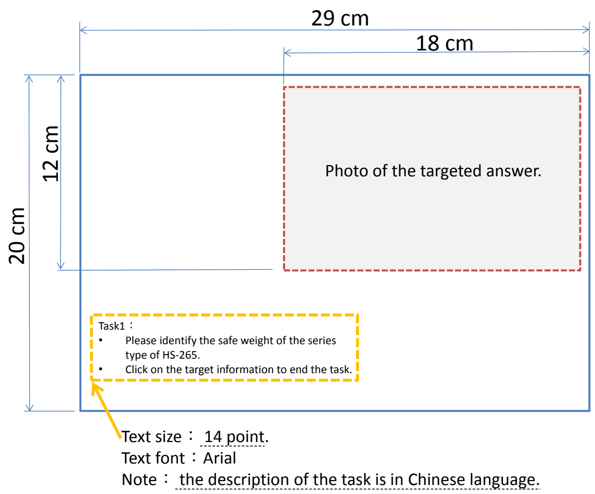


Figure 4. Size and layout design of the task card

2.6 Data analysis

Web-search performance was defined as the time spent searching, from the beginning of the task until participants found the correct answer. Each answer and task time were confirmed by the experimenter before the next task was presented. A mixed-design MANOVA was conducted to analyze the main effects and interactions of the variables. Post hoc comparisons were performed to identify the simple effects of the various factors. The statistical analyses were performed using the Statistical Package for the Social Sciences (SPSS 20.0).

3. Results

Table 2 shows descriptive statistics of the collected experimental data. Table 3 further summarizes the results of the MANOVA analysis. The variables of topological structure (F (1, 47) = 23.42, p < 0.001) and task type (F (3, 141) = 253.23, p < 0.001), but not gender (F (1, 47) = 3.21, p > 0.05), yielded significant main effects. With respect to the main effect of topological structure, the mixed structure was found to result in superior performance, consuming less time relative to the linear topological structure. With regard to task type, performance as measured by time revealed an order of easy < moderate < difficult.

There were two significant two-way interactions: topological structure × task type (F (3, 141) = 21.89, p < 0.001) and gender × task type (F (3, 141) = 3.65, p < 0.05). However, the two-way interaction between topological structure and gender (F (1, 47) = 0.114, p > 0.05), and the three-way interaction among topological structure, gender, and task type (F (3, 141) = 0.058, p > 0.05) were not significant.

Table 2. Mean time and Standard Deviation of the Task operation (s)

Variables	Levels	No.	Mean	(SD)
Topology (A)	Linear	25	92.79	(5.11)
	Mixed	26	58.15	(5.01)
Gender (B)	Female	26	81.89	(5.01)
	Male	25	69.06	(5.11)
Task type (C)	Easy	51	20.96	(1.37)
	Moderate	51	86.53	(4.93)
	Difficult	51	159.70	(8.09)

Table 3. Summary of MANOVA analysis

	Variables	Type III Sum of Squares	df.	Mean Square	*F*	Sig.	Post hoc comparison
Between-subject	Topology (A)	61129.96	1	61129.96	23.42	0.000	Mixed < Linear
	Gender (B)	8381.38	1	8381.38	3.21	0.080	
	A*B	298.29	1	298.29	0.11	0.737	
Within-subject	Task type (C)	603667.43	3	201222.48	253.23	0.000	Easy < moderate < difficult
	A*C	52185.55	3	17395.18	21.89	0.000	
	B*C	8701.36	3	2900.45	3.65	0.014	
	A*B*C	137.59	3	45.86	0.06	0.982	

(unit: second)

4. Discussion

4.1 Main effects

A significant effect of topology was found, with the mixed type yielding superior performance relative to the linear interface design. The present study not only confirmed previous findings of the topology effect, but also extended this effect by using a mixed interface design to replace the hierarchal and network topological interface designs while integrating their advantages. Hypothesis 1, which stated that a topological structure would significantly affect task performance by seniors during a web browsing task, was supported by the present study.

The experimental data analysis showed that the main effect of gender was not significant, which contrasts with several previous studies reporting a significant effect of gender (Bosco et al., 2004). However, it is important to note that the interaction between gender and task type was significant; further discussion related to this effect will be taken up in the following section.

A significant difference in performance was found across task types. Pair-wise comparisons revealed differences among task types with respect to search times, as follows: easy < moderate < difficult. This result is consistent with a previous study reporting that the greater the task complexity was, the more time it consumed (Coluccia & Louse, 2004). It is important to note that the interaction between task type and topology reached the level of statistical significance; further discussion will be devoted to identifying the nature of this interactive effect.

4.2 Interaction effects

As indicated by post hoc comparisons based on the two-way interaction of gender and task type, no significant gender difference was found in the easy (single-layer) task condition, but men performed better (taking less time) than women under moderate and difficult (double- and triple-layer) task conditions (Fig. 5). This outcome is consistent with most studies, which reported significant differences between men and women only under difficult conditions (Bosco et al., 2004).

Post hoc comparisons based on the interaction between topological structure and task type revealed no significant difference between the linear and mixed structure for the easy (single-layer) task condition, but did indicate that the mixed structure yielded superior performance relative to the linear structure under the moderate and difficult (double- and triple-layer) task conditions (Fig. 6). This result differs from a previous study that found better performance by senior citizens under a linear topological structure condition (Lin, 2004). There are two possible explanations for this discrepancy. The first is that the earlier study did not include task type as a factor. A similar point was made by Chang & Chien (2015). An easy task may be unable to reveal critical differences due to a topological structure (Coluccia et al., 2007). Another possible explanation is that the mixed topological structure integrates the merits of both a hierarchical and a network structure, thus resulting in better performance.

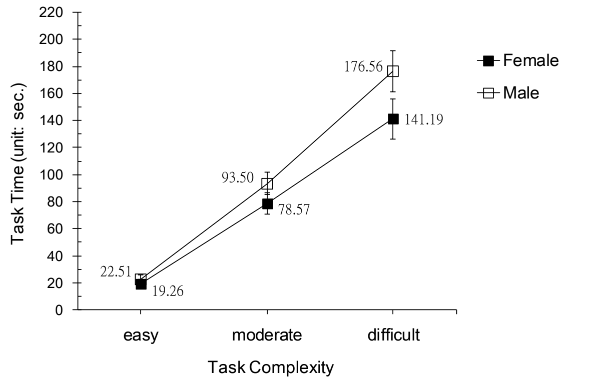


Figure 5. Mean task times by gender in each task type

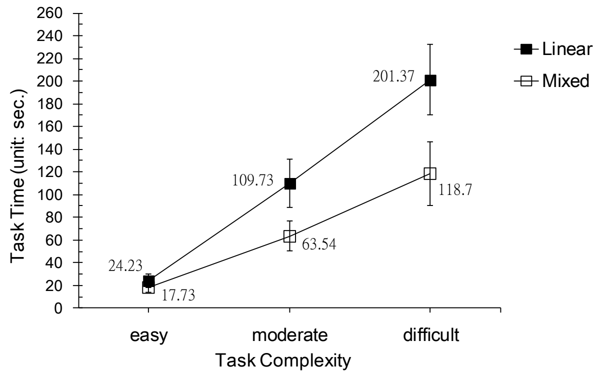


Figure 6. Mean task time by topological structure for each task type

We extended earlier examinations of interface design by proposing a mixed design to integrate the advantages of hierarchical and network topological structures, and we demonstrated the validity of this integrated design in a sample of senior citizens. We observed an interactive pattern in which a positive ratio effect was obtained between topology and task factors. Based on these findings, Hypothesis 2, which stated that gender, task type, and topological structure would have interactive effects on website browsing performance, was supported by the present study.

The present study did not include other age groups (Lindenberger, et al., 2008) or investigate wayfinding strategies (Lawton, 1996) to determine whether these would affect web browsing performance. Thus, we do not know whether younger participants would show a similar pattern to the one we observed among senior citizens, nor do we know the effects of wayfinding strategies. We recommend that future studies investigate other possible variables, including the aforementioned ones, as these may also affect the effectiveness and usability of particular website designs.

5. Conclusions

The present study provides evidence of two two-way interactions, namely an interaction between gender and task type and an interaction between topological structure and task type, in the web browsing performance of senior citizens. We also demonstrated that the mixed type of interface design, which integrates features of other topological structures (i.e., hierarchical and network structure) may enhance the efficiency of the original topological structure and result in better web browsing performance on tasks of high complexity. These results may prove beneficial for interface design and related research with older populations.

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拓撲、性別和任務類型對老年人網絡瀏覽績效的影響

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摘要

檢索具有復雜佈局的網站信息是很困難，這可能會導致錯誤的發生及沮喪感。特別是針對老年人的網頁設計而言。本研究的目為：1、確認拓撲結構，性別，任務類型對老年人網頁瀏覽任務的影響關係；2、設計和比較不同的老年人網站設計。本研究共招募 51 名年齡在 61 歲到 71 歲之間的高齡長者（26 名女性，25 名男性），來進行超文本瀏覽測試。本研究採用一個 2x2x3 混合因子實驗設計，受試者間的因子包括拓撲結構（線性和混合兩個水準）和性別（男性及女性），受試者內的因子為任務類型（分別為容易，中等和困難等 3 個水準）。結果產生了兩個二因子交互作用。首先，拓撲結構與任務類型之間的相互作用是顯著的，在中等和困難的任務條件下混合結構比線性結構有較優越的性能，但在簡單的任務中沒有差異。其次，性別與任務類型的交互作用也是顯著的，與其他二因子交互作用相比，在簡單的任務條件下沒有發現任何差異，但在適中和困難的任務條件下，男性比女性花費更少的搜索時間。在本研究中發現性別，拓撲結構，任務類型等因子對老年人超文本瀏覽性能的交互作用效果，有別於以往的研究結果。故建議在針對老年人的網頁設計時應考慮性別，拓撲結構和任務類型等因子，以增加網頁設計的價值。

關鍵詞：網站設計、老年人、任務績效

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